MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG4SECECT204 Multimedia Electronics

Duration: 1.5hours **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability.*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	Identify the parameter of sound that determines its loudness. (a) Frequency (b) Timbre (c) Amplitude (d) Overtone	U	1
2	Select the sound characteristic that differentiates two sounds of the same pitch and loudness. (a) Amplitude (b) Frequency (c) Timbre (d) Interval	U	1
3	Recognize the audio format that uses lossy compression. (a) WAV (b) FLAC (c) MP3 (d) AIFF	K	1
4	Identify the commonly used container format for online video streaming. (a) AVI (b) MKV (c) MP4 (d) MOV	K	1
5	Select the amplifier stage that boosts weak audio signals. (a) Power amplifier (b) Pre-amplifier (c) Oscillator (d) Mixer	U	1
6	Recall the connector commonly used in professional microphones. (a) RCA (b) TRS (c) XLR (d) MIDI	K	1
7	Identify the colour code of the right audio channel in RCA connectors. (a) White (b) Yellow (c) Red (d) Blue	K	1
8	Indicate the interface mainly used for digital audio transmission. (a) MIDI (b) SPDIF (c) TRS (d) RCA	U	1
9	Identify the connector used for balanced audio transmission. (a) RCA (b) TRS (c) XLR (d) MIDI	K	1
10	Recognize the amplifier class known for high efficiency. (a) Class A (b) Class B (c) Class AB (d) Class D	K	1
11	Select the hardware component crucial for video editing performance. (a) Sound card (b) Graphics card (c) Network card (d) Modem	A	2
12	Identify the diagram that represents microphone directionality. (a) Block diagram (b) Polar diagram (c) Circuit diagram (d) Timing diagram	K	2

13	Identify the loudspeaker type commonly used for low frequencies. (a) Tweeter (b) Horn (c) Woofer (d) Piezo	K	2
14	Recognize the principle used in moving-coil loudspeakers. (a) Electrostatic force (b) Magnetic induction (c) Piezoelectric effect (d) Photoelectric effect	U	2
15	Select the advantage of fibre optic cable for video transmission. (a) Low bandwidth (b) High attenuation (c) Immunity to EMI (d) Low speed	U	2
16	Identify the microphone best suited for stage performances. (a) Ribbon (b) Condenser (c) Dynamic (d) Carbon	A	2
17	Recognize the video cable capable of carrying both audio and video. (a) DVI (b) HDMI (c) VGA (d) Coaxial	K	2
18	Identify the technology used for replacing a background in video production. (a) Live streaming (b) Encoding (c) Chroma key (d) Rendering	U	2
19	Recognize the process of broadcasting video content in real time. (a) Encoding (b) Rendering (c) Live streaming (d) Editing	U	2
20	Indicate the microphone characteristic that affects sensitivity. (a) Frequency response (b) Polar pattern (c) Impedance (d) All of these	U	2
21	Identify the function of a crossover network. (a) Amplification (b) Frequency separation (c) Mixing signals (d) Recording audio	U	3
22	Recognize the device used to combine multiple audio signals. (a) Equalizer (b) Mixer console (c) Amplifier (d) Recorder	K	3
23	Select the control that adjusts specific frequency bands. (a) Tone control (b) Graphic equalizer (c) Pre-amplifier (d) Delay unit	U	3
24	Identify the function of a graphic equalizer. (a) Amplification (b) Frequency band adjustment (c) Signal generation (d) Recording	U	3
25	Select the circuit that modifies bass and treble levels. (a) Oscillator (b) Tone control circuit (c) Rectifier (d) Modulator	U	3
26	Recognize the advantage of wireless headphones over wired ones. (a) Higher fidelity (b) No battery usage (c) Mobility (d) Lower latency	U	3
27	Identify the role of a tweeter in a speaker system. (a) Low-frequency output (b) Mid-range output (c) High-frequency output (d) Noise cancellation	K	3
28	Recognize a device commonly used by DJs for live mixing. (a) Audio interface (b) DJ console (c) Power amplifier (d) Speaker	K	4
29	Identify the purpose of special audio effects. (a) Signal attenuation (b) Creative enhancement (c) Noise removal only (d) Power boosting	U	4
30	Recognize the output obtained after rendering a video project. (a) Project file (b) Raw clip (c) Final video file (d) Timeline	U	4
31	Select the software used for audio editing.	U	4

	(a) Audacity (b) Blender (c) MATLAB (d) AutoCAD		
32	Identify the device that introduces time delay in audio. (a) Equalizer (b) Delay unit (c) Compressor (d) Limiter	K	4
33	Identify the software suitable for basic video editing. (a) Nero Wave Editor (b) Audacity (c) Filmora (d) MS Word	U	4
34	Synchronizing audio with video mainly employs . (a) Matching colour depth (b) Aligning audio track with video timeline (c) Reducing file size (d) Adding subtitles	A	4
35	Choose the first step in making a short movie. (a) Rendering (b) Editing (c) Recording (d) Exporting	An	4

1 * 35 = 35

Part B

Short Answer Questions

Answer any **3** Questions

Each question carries 5 marks

36	Explain the characteristics and properties of sound such as amplitude, frequency, pitch, timbre and harmonics.	U	1
37	Describe video formats and interfaces such as HDMI and DVI, and explain their applications.	U	2
38	Classify different types of loudspeakers and explain their working principles.	An	3
39	Explain the working and applications of crossover networks and public address systems.	U	3
40	Describe the steps involved in creating a short movie using audio and video editing software.	U	4

5 * 3 = 15